

Setare Hami

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Education

Sharif University of Technology

B.Sc. in Mathematics, Minor in Computer Science

GPA: 18.52/20

Tehran, Iran

Sept. 2022–Expected Feb. 2027

Selected Coursework

Elementary Algebraic Geometry	20/20	Homological Algebra ¹	20/20
Algebra I	19.1/20	Topology & Geometry Seminar	20/20
Differentiable Manifolds (Graduate)	20/20	General Topology	20/20
Differential Topology	19.5/20	Linear Algebra I	19.1/20
Group Theory ²	In Progress	Topics in Algebraic Geometry (Graduate)	In Progress

¹Listed as “Number Theory” on the official transcript.

²Topics include finite groups, Lie groups, and representations of groups and Lie algebras.

Academic Experience

B.Sc. Project – Group Schemes

2026–Present

Supervised by Dr. Mohammad Hadi Hedayatzaadeh, IPM

- Undertaking a supervised reading project on group schemes, based on Waterhouse’s *Introduction to Affine Group Schemes*.

Commutative Algebra Course

2026–2027

Institute for Research in Fundamental Sciences (IPM), Tehran, Iran

Instructor: Dr. Mohammad Hadi Hedayatzaadeh

- Studied commutative algebra based on *The CRing Project*, covering fields and extensions, localization, tensor products and Hom, spectra of rings, Noetherian rings and modules, graded and filtered rings, and introductory topics in integrality.

Talks & Presentations

Quotients in Algebraic Geometry: From Categorical Quotients to Schemes, Algebraic Spaces, and Quotient Stacks

Sep. 2026

Schemeathon – IPM Algebraic Geometry Immersion Program

Co-presented a two-person project developed during the Schemeathon.

Lyndon–Hochschild–Serre Spectral Sequence

Aug. 2026

Homological Algebra, Sharif University of Technology

Presented the construction and convergence of the LHS spectral sequence for group cohomology via the Grothendieck spectral sequence, including a low-degree consequence and an explicit non-split group extension.

Geometric Group Theory

Spring 2026

Topology & Geometry Seminar, Sharif University of Technology

Co-presented material from Chapters 1–6 of Clara Löh’s *Geometric Group Theory: An Introduction*.

Schools & Workshops

Schemeathon – IPM Algebraic Geometry Immersion Program

Aug. 2026

Khansar Campus, University of Isfahan, Iran

Seven-day residential program in algebraic geometry emphasizing problem-solving, collaborative work, and mini-projects.

Third Winter School on Algebraic Geometry and Number Theory

Feb. 2026

Institute for Research in Fundamental Sciences (IPM), Isfahan, Iran

Topics: Artin–Schreier and Kummer Theory; Symmetric Spaces and their Cohomology; Tannakian Formalism.
Second Winter School on Algebraic Geometry and Number Theory Feb. 2025
Institute for Research in Fundamental Sciences (IPM), Isfahan, Iran
Courses: Algebraic Curves and Surfaces; p -adic and ℓ -adic Galois Representations; Simplicial Methods in Algebraic Geometry and Number Theory.
Ninth Summer School of Mathematics Aug. 2024
Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran
Courses: Analytic Number Theory; Sharp Threshold Phenomena in Probability.

Teaching Experience

Teaching Assistant, Sharif University of Technology			
Algebra	Spring 2025	Linear Algebra	Fall 2026
Topology	Spring 2025, 2026	Complex Analysis	Fall 2026
Mathematical Analysis I & II	Spring 2025, 2026	Elementary Functional Analysis	Fall 2026
Introduction to Graph Theory and Applications	Spring 2024	Basic Programming (Java)	Fall 2023
Mathematics Olympiad Instructor		Sept. 2022–June 2023; Sept. 2024–June 2025	

Honors & Awards

Bronze Medal, Iranian Mathematical Olympiad	2021
Ranked 4th among 28 students, Department of Mathematics, Sharif University of Technology	
Top 0.4% nationwide (569/145,000), Iranian University Entrance Examination	2022
Third Place, 12th Kalaghha Mathematics Competition	2024
Second Place, Code Knock Algorithmic Competition, Sharif University of Technology	2023

Academic Service

Member, Education Committee, Faculty Student Council, Sharif University of Technology	Dec. 2024–Present
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Technical Skills

Programming & Computing: Python, Java, MATLAB, Maxima

Languages

Persian: Native	
English: IELTS Academic 7.5 (R 8.0, L 8.0, S 7.0, W 7.0),	Nov. 2025